

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Unit Test 1

Unit Test 1 - Answer Key

Class: Class 10

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Networking Concepts

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: MCQ (5 x 1 = 5 marks)

1. **Answer:** b - Star topology
 - **Explanation:** Star topology uses a central hub or switch to connect all devices. All data passes through the central node.
2. **Answer:** c - 7
 - **Explanation:** The OSI (Open Systems Interconnection) model has 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
3. **Answer:** b - 172.16.0.1
 - **Explanation:** Class B IP addresses range from 128.0.0.0 to 191.255.255.255. 172.16.0.1 falls in this range. 192.168.x.x is Class C, 10.x.x.x is Class A.
4. **Answer:** c - SMTP
 - **Explanation:** SMTP (Simple Mail Transfer Protocol) is used to send emails. POP3 and IMAP are used to receive emails. HTTP is for web browsing, FTP for file transfer.
5. **Answer:** b - To filter incoming and outgoing network traffic
 - **Explanation:** A firewall monitors and controls network traffic based on security rules, blocking unauthorized access while allowing legitimate communication.

Section II: Short Answer (5 x 1 = 5 marks)

1. **IPv4 vs IPv6**
 - **Answer:** IPv4 uses 32-bit addresses (e.g., 192.168.1.1), providing 4 billion addresses. IPv6 uses 128-bit addresses (e.g., 2001:0db8:85a3::8a2e:0370:7334), providing virtually unlimited addresses. IPv6 was introduced because IPv4 addresses are exhausted. IPv6 also has better security and routing.
 - **Marking:** 1 mark for comparison, 1 mark for why needed.
2. **Star Topology diagram**
 - **Answer:** Diagram should show a central hub/switch with devices connected to it like spokes. Advantages: Easy to install, failure of one device doesn't affect others. Disadvantages: If hub fails, entire network goes down; requires more cable than bus topology.
 - **Marking:** 1 mark for diagram, 0.5 marks for each advantage/disadvantage.
3. **Full forms and purposes**
 - **Answer:** HTTP (HyperText Transfer Protocol) - transfers web pages. FTP (File Transfer Protocol) - transfers files between computers. SMTP (Simple Mail Transfer Protocol) - sends emails. DNS (Domain Name System) - translates domain names to IP addresses. TCP (Transmission Control Protocol) - ensures reliable data delivery.
 - **Marking:** 0.2 marks for each full form and purpose.
4. **MAC address vs IP address**

- **Answer:** MAC address is a hardware address burned into the network card (e.g., 00:1A:2B:3C:4D:5E). IP address is a logical address assigned by the network (e.g., 192.168.1.1). MAC is permanent, IP changes with network. MAC works at Data Link layer, IP works at Network layer.
- **Marking:** 1 mark for each comparison point.

5. Data travel through Internet

- **Answer:** When you send data: (1) Your computer creates packets with source/destination IP. (2) Packets travel through NIC to router. (3) Router examines IP and forwards to next hop using routing table. (4) Packets traverse multiple routers and ISPs. (5) At destination, packets are reassembled. Protocols: TCP ensures reliable delivery, IP handles addressing, HTTP/FTP for application data. Devices: NIC, router, switch, modem.
- **Marking:** 1 mark for process, with at least 3 devices/protocols mentioned.